

Supervised Learning - Model Performance Evaluation

100 Questions: Continuous Metrics, Classification Metrics, ROC/AUC, Threshold Selection

GARP RAI Exam Preparation

Q1: Model Evaluation Purpose - Tutorial

QUESTION: What is the primary purpose of model performance evaluation in supervised learning?

OPTIONS:

- ❶ To increase the complexity of the model
- ❷ To assess how well a model generalizes to new, unseen data and compare different models
- ❸ To add more features to the model
- ❹ To prepare the data for training

ANSWER: B

DETAILED TUTORIAL:

- **Primary purposes of model evaluation:**
 - ❶ **Generalization assessment:** How well does the model perform on new data?
 - ❷ **Model comparison:** Which model is best for the specific problem?
 - ❸ **Error quantification:** How wrong can we afford to be?
 - ❹ **Risk management:** What are the consequences when models are wrong?
- **Why it matters in risk management:**
 - Models drive critical decisions (credit, fraud, AML)
 - Poor models have significant consequences
 - Regulatory requirements demand validation
 - Stakeholder confidence depends on reliable models

EXAM TRAP: Model evaluation is about **generalization performance**, not training performance!

Q2: Continuous vs Classification - Tutorial

QUESTION: Why is a distinction made between evaluation metrics for continuous vs discrete outputs?

OPTIONS:

- ① Because continuous outputs don't need evaluation
- ② Different types of outputs require different types of error measurement and interpretation
- ③ Because classification models are always better
- ④ Because continuous models are always more complex

ANSWER: B

DETAILED TUTORIAL:

- **Continuous outputs (regression):**
 - Predict numbers: VaR, prices, revenues, losses
 - Need measures of numerical accuracy
 - Error magnitude matters
 - Examples: MSE, RMSE, MAE, MAPE
- **Discrete outputs (classification):**
 - Predict categories: fraud vs legitimate, default vs paid
 - Need measures of classification accuracy
 - Type of error matters (false positives vs false negatives)
 - Examples: Accuracy, Precision, Recall, F1, AUC

EXAM TRAP: Continuous vs discrete outputs require fundamentally different evaluation approaches!

Q3: Poor Modeling Consequences - Tutorial

QUESTION: What are the potential consequences of poor model performance in banking?

OPTIONS:

- ① Reduced capital requirements
- ② Capital add-ons, shareholder dilution, regulatory scrutiny, and reputational damage
- ③ Decreased regulatory oversight
- ④ Improved market confidence

ANSWER: B

DETAILED TUTORIAL:

- **Example from the chapter:**
 - Greek bank estimated recovery rate at 65%
 - Actual recovery was 40%
 - 25-point overestimate forced €120 million capital add-on
 - Triggered stock price slide
- **Chain of consequences:**
 - ① **Financial impact:** Capital add-ons, shareholder dilution
 - ② **Operational impact:** Deep-dive reviews, resource consumption
 - ③ **Strategic impact:** Revealed systemic weaknesses
 - ④ **Reputational impact:** Regulator and market confidence erosion

EXAM TRAP: Poor modeling has **real financial** and **reputational** consequences!

Q4: Test Sample Importance - Tutorial

QUESTION: Why is it important to compute performance metrics on a test sample rather than the training sample?

OPTIONS:

- ❶ Because the test sample is larger
- ❷ Models often overfit to training data, so training performance is not a true test of generalization
- ❸ Because test samples are easier to compute
- ❹ Because training samples contain errors

ANSWER: B

DETAILED TUTORIAL:

- **Overfitting problem:**
 - Models learn training data patterns
 - Including noise and random fluctuations
 - Performance on training data is optimistic
 - Does not reflect true generalization
- **Data split purposes:**
 - **Training sample:** Estimate parameters
 - **Validation sample:** Tune hyperparameters
 - **Test sample:** Evaluate final performance

EXAM TRAP: Always evaluate on **test sample** for true generalization performance!

Q5: Training vs Test Gap - Tutorial

QUESTION: What does a significant gap between training and test performance indicate?

OPTIONS:

- ❶ The model is performing perfectly
- ❷ The model is likely overfitting to the training data
- ❸ The model is underfitting
- ❹ The data is perfectly clean

ANSWER: B

DETAILED TUTORIAL:

- **Performance patterns:**
 - **No gap:** Training equal to Test (good generalization)
 - **Small gap:** Slight overfitting (acceptable)
 - **Large gap:** Significant overfitting (problematic)
- **Why gap occurs:**
 - Model memorizes training patterns
 - Learns noise instead of signal
 - Too complex for the data
 - Insufficient regularization

EXAM TRAP: Large train-test performance gap = **overfitting!**

Q6: MSE Definition - Tutorial

QUESTION: What is Mean Squared Error (MSE) and what does it measure?

OPTIONS:

- ① Average of absolute differences between predictions and actual values
- ② Average of squared differences between predicted and actual values
- ③ Percentage of correctly classified instances
- ④ Average of squared differences divided by actual values

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

- $MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$
- y_i : true value
- $\hat{y}_i$: predicted value
- N : number of observations

• **What it measures:**

- Average squared deviation
- Penalizes large errors more heavily
- Sensitive to outliers

• **Key characteristics:**

- Always non-negative
- Lower is better
- Units are squared (e.g., dollars squared)
- Heavily penalizes large deviations

EXAM TRAP: MSE uses **squared errors** - heavily penalizes large deviations!

QUESTION: What managerial question does MSE help answer?

OPTIONS:

- 1 "What does the typical error feel like in our primary unit of currency?"
- 2 "How do we penalize big errors in our model?"
- 3 "On average, by how much are our predictions off, regardless of direction?"
- 4 "In percentage terms, what is the typical error relative to the actual value?"

ANSWER: B

DETAILED TUTORIAL:

- **Managerial focus:**
 - Highlights tail risk
 - Emphasis on squared errors heavily penalizes large deviations
 - Sensitive to costly outliers
 - Important for risk management
- **Why this matters:**
 - Large errors have significant consequences
 - Risk managers care about tail events
 - Regulatory focus on extreme outcomes

EXAM TRAP: MSE is for **penalizing big errors** - risk managers love it!

Q8: RMSE Definition - Tutorial

QUESTION: What is Root Mean Square Error (RMSE) and how does it differ from MSE?

OPTIONS:

- ① RMSE is the average of absolute errors
- ② Square root of MSE; scales with units of data, easier to interpret
- ③ RMSE is always larger than MSE
- ④ RMSE is the percentage of errors

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

$$• RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

• **Key characteristics:**

- Same units as target variable (e.g., dollars)
- More interpretable than MSE
- Gives sense of average error magnitude
- Still sensitive to outliers

• **Comparison to MSE:**

- MSE: dollars squared (hard to interpret)
- RMSE: dollars (easy to interpret)
- Both penalize large errors

EXAM TRAP: RMSE is **easier to interpret** than MSE because it's in the same units!

QUESTION: What managerial question does RMSE help answer?

OPTIONS:

- ❶ "How do we penalize big errors in our model?"
- ❷ "What does the typical error feel like in our primary unit of currency?"
- ❸ "On average, by how much are our predictions off?"
- ❹ "In percentage terms, what is the typical error?"

ANSWER: B

DETAILED TUTORIAL:

- **Managerial focus:**
 - Same units as target variable (e.g., dollars)
 - Directly interpretable
 - "On average, our predictions are off by RMSE dollars"
 - Helps non-technical stakeholders understand
- **Caveat:**
 - Still sensitive to outliers
 - Unusually large errors can inflate RMSE
 - Need to investigate if high RMSE is systemic or from extreme events

EXAM TRAP: RMSE answers "what does the typical error feel like?"

Q10: MAE Definition - Tutorial

QUESTION: What is Mean Absolute Error (MAE) and what is its key advantage?

OPTIONS:

- ① Average of squared errors; penalizes large errors
- ② Average of absolute differences; robust to outliers
- ③ Average of percentage errors; good for comparisons
- ④ Average of errors; can cancel out positive and negative

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

• **Key characteristics:**

- Uses absolute differences (not squared)
- Robust to outliers
- Easy to communicate
- Straightforward average error magnitude

• **Limitations:**

- Can mask direction of bias
- Doesn't penalize occasional large misses
- May underestimate critical risk events

EXAM TRAP: MAE is **robust to outliers** - unlike MSE and RMSE!

QUESTION: What managerial question does MAE help answer?

OPTIONS:

- ① "How do we penalize big errors?"
- ② "What does the typical error feel like?"
- ③ "On average, by how much are our predictions off, regardless of direction?"
- ④ "In percentage terms, what is the typical error?"

ANSWER: C

DETAILED TUTORIAL:

- **Managerial focus:**
 - "On average, our predictions are off by MAE dollars"
 - Ignores direction (over vs under prediction)
 - Easy to communicate to non-technical audiences
- **When to use:**
 - When outliers are not critical
 - When you want stable, interpretable measure
 - When equal treatment of all errors is desired

EXAM TRAP: MAE answers "by how much are we off, regardless of direction?"

Q12: MAPE Definition - Tutorial

QUESTION: What is Mean Absolute Percentage Error (MAPE) and what is its key advantage?

OPTIONS:

- ① Average of squared percentage errors
- ② Average of absolute percentage errors; provides normalized percentage measure
- ③ Average of percentage errors; can cancel out
- ④ Average of absolute errors divided by predictions

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

$$• \text{MAPE} = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|$$

• **Key characteristics:**

- Expressed as percentage
- Normalized - allows comparison across different scales
- Board-friendly (percentage nature)
- Consistent across portfolios

• **Limitations:**

- Explodes when actual values are zero or near zero
- Can be misleading with small denominators
- May understate errors on large absolute value items

EXAM TRAP: MAPE is **percentage-based** - great for communication but has limitations!

QUESTION: What managerial question does MAPE help answer?

OPTIONS:

- ❶ "How do we penalize big errors?"
- ❷ "What does the typical error feel like?"
- ❸ "On average, by how much are our predictions off?"
- ❹ "In percentage terms, what is the typical error relative to the actual value?"

ANSWER: D

DETAILED TUTORIAL:

- **Managerial focus:**
 - "On average, our predictions are off by MAPE%"
 - Normalized - allows comparison across models/portfolios
 - Perfect for audit committees and board presentations
 - Common choice for auditors
- **When to use:**
 - Comparing models with different scales
 - Communicating to non-technical stakeholders
 - When percentage errors are more meaningful than absolute

EXAM TRAP: MAPE answers "what's the error as a percentage?"

Q14: MSE Calculation Example - Tutorial

QUESTION: Given a test sample of 10 observations, how is MSE calculated?

OPTIONS:

- ❶ Sum of absolute errors divided by N
- ❷ Sum of squared errors divided by N
- ❸ Square root of sum of squared errors divided by N
- ❹ Sum of errors divided by N

ANSWER: B

DETAILED TUTORIAL:

• **Step-by-step with example:**

- ❶ Predict values using the model
- ❷ Calculate errors: $y_i - \hat{y}_i$
- ❸ Square each error: $(y_i - \hat{y}_i)^2$
- ❹ Sum all squared errors
- ❺ Divide by N (number of observations)

• **Example from chapter:**

- Salary prediction model: $\hat{y} = 16.88 + 1.06 \times \text{experience}$
- Sum of squared errors: 431.02
- $N = 10$
- $\text{MSE} = 431.02 / 10 = 43.1$
- $\text{RMSE} = \sqrt{43.1} = 6.57$

EXAM TRAP: MSE = sum of squared errors divided by N!

QUESTION: In the salary prediction example with $RMSE = 6.57$, what does this mean?

OPTIONS:

- ❶ The model is off by exactly 6.57 for every prediction
- ❷ On average, the model's predictions are off by about 6.57 units
- ❸ The model is 6.57% accurate
- ❹ The model has 6.57% error rate

ANSWER: B

DETAILED TUTORIAL:

• **Interpretation:**

- "On average, predictions are off by about 6.57 thousand dollars"
- RMSE is in same units as the target (salary in \$000s)
- Gives sense of typical error magnitude

• **Important nuances:**

- This is an average - some errors are larger, some smaller
- Squaring means larger errors contribute more to the average
- RMSE eq 6.57 means typical error is around 6.57 units

EXAM TRAP: RMSE is in **same units as target** - easier to understand!

Q16: MAE Example Calculation - Tutorial

QUESTION: In the salary prediction example, $MAE = 5.01$. What does this mean and how is it calculated?

OPTIONS:

- ① Average of squared errors = 5.01
- ② Average of absolute errors = 5.01 (sum of absolute errors divided by N)
- ③ Average of percentage errors = 5.01%
- ④ Square root of MSE = 5.01

ANSWER: B

DETAILED TUTORIAL:

• **Calculation:**

- Sum of absolute errors = 50.11 (from Table 8.3)
- $N = 10$
- $MAE = 50.11 / 10 = 5.01$

• **Interpretation:**

- "On average, predictions are off by about 5.01 thousand dollars"
- Similar to RMSE but less influenced by large errors
- More stable measure of typical error

EXAM TRAP: $MAE = \text{average of absolute errors}$ - sum divided by N!

QUESTION: In the salary example, $\text{MAPE} = 16\%$. What does this mean?

OPTIONS:

- ❶ The model is 16% accurate
- ❷ The average forecast error is 16% of actual salary
- ❸ The model makes errors 16% of the time
- ❹ The model has 16% bias

ANSWER: B

DETAILED TUTORIAL:

• **Calculation:**

- Sum of $|(y_i - \hat{y}_i)/y_i| = 1.62$
- $N = 10$
- $\text{MAPE} = 1.62 / 10 \times 100 = 16.2\%$

• **Interpretation:**

- "On average, the forecast error is 16% of the actual salary"
- Most intuitive of all measures
- Easy to communicate to non-technical stakeholders

EXAM TRAP: $\text{MAPE} = \text{percentage of actual values}$ - most intuitive for communication!

QUESTION: Which metric would a risk manager choose to capture tail risk?

OPTIONS:

- 1 MAE
- 2 MSE or RMSE
- 3 MAPE
- 4 Accuracy

ANSWER: B

DETAILED TUTORIAL:

- **Why MSE/RMSE capture tail risk:**
 - Squaring errors heavily penalizes large deviations
 - Large errors contribute disproportionately
 - Sensitive to outliers (tail events)
 - Regulatory preference (ECB recommends)
- **Comparison:**
 - MAE: Robust, but ignores tail risk
 - MAPE: Good for communication, but has limitations
 - MSE/RMSE: Best for capturing extreme errors

EXAM TRAP: ECB recommends metrics that **capture tail risk** - MSE and RMSE!

QUESTION: When should a risk manager use MAPE instead of MSE or RMSE?

OPTIONS:

- ① When comparing models with very different scales
- ② When tail risk is the primary concern
- ③ When errors are always very small
- ④ When the model is overfitting

ANSWER: A

DETAILED TUTORIAL:

- **When MAPE is appropriate:**
 - Comparing models across different portfolios
 - Comparing a €100k loan vs €10m loan
 - Communicating to audit committees
 - When percentage errors are more meaningful
- **When MAPE is NOT appropriate:**
 - When actual values are near zero
 - When tail risk is the primary concern
 - When absolute errors matter more than percentage

EXAM TRAP: MAPE is best for **comparing across different scales!**

QUESTION: Why should risk managers look beyond average metrics and examine error distributions?

OPTIONS:

- ❶ Average metrics are always sufficient
- ❷ The distribution can reveal patterns or biases that averages might hide
- ❸ Error distributions are easier to calculate
- ❹ Averages are not used in risk management

ANSWER: B

DETAILED TUTORIAL:

- ❶ Key questions to ask:
 - "Are errors generally small with a few large ones?"
 - "Are errors randomly distributed or systematically biased?"
 - "Are there segments where errors are consistently larger?"
- ❷ Why this matters:
 - Consistent bias $\neq$ random errors
 - Segments with larger errors need attention
 - Visualizations (histograms, box plots) provide richer context

EXAM TRAP: Error distributions reveal **patterns and biases** that averages hide!

QUESTION: In the house price prediction example, how do we determine which model is better?

OPTIONS:

- ❶ Always choose the more complex model
- ❷ Compare performance metrics on test sample - lower values indicate better performance
- ❸ Compare performance metrics on training sample
- ❹ Always choose the simpler model

ANSWER: B

DETAILED TUTORIAL:

• **Example results (Table 8.4):**

- Model A (linear regression): $MSE=94.85$, $MAE=6.56$, $MAPE=18.25$
- Model B (decision tree): $MSE=77.35^*$, $MAE=5.41^*$, $MAPE=14.45^*$

• **Conclusion:**

- Model B has all lower values
- Decision tree is better according to all metrics
- Asterisk denotes best performer

EXAM TRAP: Lower values = **better performance** for all these metrics!

QUESTION: Can different performance metrics yield different model rankings?

OPTIONS:

- ❶ No, all metrics always give the same ranking
- ❷ Yes, different metrics can yield different rankings; choice depends on the problem
- ❸ Only MSE and RMSE give different rankings
- ❹ Only MAE and MAPE give different rankings

ANSWER: B

DETAILED TUTORIAL:

- **Why rankings can differ:**
 - MSE/RMSE: Penalize large errors
 - MAE: Treats all errors equally
 - MAPE: Emphasizes relative errors
 - Different metrics = different priorities
- **Implication:**
 - Choose metric based on business needs
 - Regulatory requirements may dictate choice
 - Historical organizational choices matter

EXAM TRAP: Different metrics = **different rankings** - choose based on problem!

QUESTION: What is a limitation of unnormalized metrics like RMSE and MAE?

OPTIONS:

- ❶ They are too difficult to calculate
- ❷ They treat errors symmetrically regardless of the scale of the target
- ❸ They always prefer complex models
- ❹ They cannot be used for risk management

ANSWER: B

DETAILED TUTORIAL:

- **The problem:**
 - \$148,000 vs \$150,000 = \$2,000 error (1)
 - \$10,000 vs \$12,000 = \$2,000 error (10)
 - Both contribute same to RMSE/MAE
 - But the second error is much more significant
- **Solution:**
 - Use MAPE for relative comparison
 - Understand the context of errors
 - Consider business impact, not just statistical measures

EXAM TRAP: Unnormalized metrics treat **all errors equally** - MAPE handles this issue!

QUESTION: What is the ECB's recommendation regarding performance metrics?

OPTIONS:

- ❶ Only use MAPE for all models
- ❷ Use metrics that capture tail risk, nudging firms to report RMSE alongside MAE
- ❸ Always use MAE because it's most robust
- ❹ Never use MSE because it's too complex

ANSWER: B

DETAILED TUTORIAL:

- **ECB Guide to Internal Models:**
 - Recommends metrics that capture tail risk
 - Nudges firms to report RMSE alongside MAE
 - Ensures large, infrequent losses are not overlooked
- **Why this matters:**
 - MAE alone might mask critical risks
 - RMSE captures the impact of extreme events
 - Both together provide complete picture

EXAM TRAP: ECB recommends **RMSE alongside MAE** for complete risk picture!

QUESTION: Why do auditors often gravitate to MAPE?

OPTIONS:

- ① Because MAPE is most accurate
- ② For its pure percentage narrative when discussing model accuracy with audit committees
- ③ Because MAPE captures tail risk best
- ④ Because MAPE is easiest to calculate

ANSWER: B

DETAILED TUTORIAL:

- **Auditor perspective:**
 - MAPE provides standardized percentage measure
 - Easy to understand for non-technical stakeholders
 - Allows comparison across different models/portfolios
 - Communicates effectively with audit committees
- **Why this matters:**
 - Different stakeholders prefer different metrics
 - Regulators vs Auditors have different needs
 - Risk managers must satisfy multiple audiences

EXAM TRAP: Auditors prefer MAPE for its **percentage narrative**!

QUESTION: What is a confusion matrix in classification model evaluation?

OPTIONS:

- ① A matrix showing model parameters
- ② A cross-tabulation of predicted classifications against actual outcomes
- ③ A matrix showing feature correlations
- ④ A matrix showing error rates

ANSWER: B

DETAILED TUTORIAL:

• **Structure:**

	Predicted Negative	Predicted Positive
Actual Negative	True Negative (TN)	False Positive (FP)
Actual Positive	False Negative (FN)	True Positive (TP)

• **Why it's important:**

- Foundation for all classification metrics
- Shows all four types of outcomes
- Enables calculation of Precision, Recall, Accuracy, F1

EXAM TRAP: Confusion matrix = **predicted vs actual** cross-tabulation!

QUESTION: What is a True Positive (TP) in a confusion matrix?

OPTIONS:

- ❶ Correctly identified a negative case
- ❷ Correctly identified a positive case
- ❸ Incorrectly identified a negative case as positive
- ❹ Incorrectly identified a positive case as negative

ANSWER: B

DETAILED TUTORIAL:

- **Definition:** Model correctly identified a positive case
- **Examples:**
 - Correctly flagged a fraudulent transaction
 - Correctly predicted that a loan would default
 - Correctly identified a dividend-paying firm
- **Desired outcome:**
 - This is what we want the model to do
 - More TP = better model performance

EXAM TRAP: True Positive = correctly identified positive!

QUESTION: What is a True Negative (TN) in a confusion matrix?

OPTIONS:

- ❶ Correctly identified a negative case
- ❷ Correctly identified a positive case
- ❸ Incorrectly identified a negative case as positive
- ❹ Incorrectly identified a positive case as negative

ANSWER: A

DETAILED TUTORIAL:

- **Definition:** Model correctly identified a negative case
- **Examples:**
 - Correctly identified a legitimate transaction
 - Correctly predicted that a loan would not default
 - Correctly identified a non-dividend paying firm
- **Desired outcome:**
 - This is what we want the model to do
 - More TN = better model performance

EXAM TRAP: True Negative = correctly identified negative!

QUESTION: What is a False Positive (FP) in a confusion matrix?

OPTIONS:

- ❶ Correctly identified a negative case
- ❷ Correctly identified a positive case
- ❸ Incorrectly classified a negative case as positive (Type I Error)
- ❹ Incorrectly classified a positive case as negative (Type II Error)

ANSWER: C

DETAILED TUTORIAL:

- **Definition:** Model incorrectly classified a negative case as positive
- **Also known as:** Type I Error
- **Examples:**
 - Flagged a legitimate transaction as fraudulent
 - Predicted that a good loan would default
 - False alarm
- **Cost perspective:**
 - Operational cost to investigate
 - Customer friction
 - Lost revenue from declined good customers

EXAM TRAP: False Positive = Type I Error - incorrectly saying "yes" !

QUESTION: What is a False Negative (FN) in a confusion matrix?

OPTIONS:

- ❶ Correctly identified a negative case
- ❷ Correctly identified a positive case
- ❸ Incorrectly classified a negative case as positive
- ❹ Incorrectly classified a positive case as negative (Type II Error)

ANSWER: D

DETAILED TUTORIAL:

- **Definition:** Model incorrectly classified a positive case as negative
- **Also known as:** Type II Error
- **Examples:**
 - Missed a fraudulent transaction (labeled legitimate)
 - Failed to predict that a loan would default
 - Missed a critical compliance breach
- **Cost perspective:**
 - Often the most costly type of error
 - Financial losses from undetected fraud
 - Capital impact of missed defaults
 - Regulatory sanctions

EXAM TRAP: False Negative = Type II Error - incorrectly saying "no"!

Q31: Type I vs Type II Errors - Tutorial

QUESTION: Which type of error is typically more costly in credit risk management?

OPTIONS:

- ❶ Type I Error (False Positive)
- ❷ Type II Error (False Negative)
- ❸ Both are equally costly
- ❹ Neither is costly

ANSWER: B

DETAILED TUTORIAL:

- **Why Type II Errors are more costly:**
 - Misclassifying a borrower as solvent who defaults
 - Results in actual loan losses
 - Regulatory capital implications
 - Direct financial impact
- **Type I Error cost:**
 - Lost profit from not making a loan
 - Operational cost of investigation
 - Customer friction
 - Usually less severe than Type II
- **General principle:**
 - The cost of errors is typically not symmetric
 - Choose threshold based on relative costs

EXAM TRAP: Type II Errors (False Negatives) are **usually more costly** in credit risk!

QUESTION: What is accuracy in classification model evaluation?

OPTIONS:

- ➊ Percentage of correctly classified positive cases
- ➋ Percentage of correctly classified instances overall
- ➌ Percentage of negative cases correctly classified
- ➍ Average of precision and recall

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

$$\bullet \text{ Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

• **Interpretation:**

- "Percentage of all predictions that were correct"
- Most intuitive classification metric
- Easy to communicate

• **Limitations:**

- Doesn't distinguish between error types
- Misleading for imbalanced datasets
- Doesn't account for costs of errors

EXAM TRAP: Accuracy = percentage of all correct predictions!

Q33: Accuracy Example - Tutorial

QUESTION: In the dividend prediction example with TP=432, TN=279, FP=121, FN=168, what is accuracy?

OPTIONS:

- ① 43.2%
- ② 71.1%
- ③ 78.1%
- ④ 28.9%

ANSWER: B

DETAILED TUTORIAL:

• **Calculation:**

- $\text{Accuracy} = (432 + 279) / (432 + 279 + 121 + 168)$
- $= 711 / 1000$
- $= 71.1\%$

• **Interpretation:**

- "71.1% of dividend predictions were correct"
- Error rate $= 1 - 0.711 = 28.9\%$

• **Context:**

- 600 firms actually paid dividends
- 400 firms actually did not
- Model correctly classified 71.1% of all firms

EXAM TRAP: Accuracy $= (TP + TN) / \text{Total!}$

QUESTION: What is the error rate and how is it related to accuracy?

OPTIONS:

- ❶ Error rate = 1 - Accuracy
- ❷ Error rate = Accuracy + 1
- ❸ Error rate = Accuracy / 2
- ❹ Error rate = 2 × Accuracy

ANSWER: A

DETAILED TUTORIAL:

• **Formula:**

$$\bullet \text{ ErrorRate} = 1 - \text{Accuracy} = \frac{FP+FN}{TP+TN+FP+FN}$$

• **Interpretation:**

- "Percentage of predictions that were incorrect"
- Complementary to accuracy
- Simple and intuitive

• **Example:**

- Accuracy = 71.1%
- Error Rate = 100% - 71.1% = 28.9%
- "28.9% of predictions were wrong"

EXAM TRAP: Error rate = 1 - Accuracy!

QUESTION: What is a major limitation of accuracy as a performance measure?

OPTIONS:

- ❶ It's too difficult to calculate
- ❷ It fails to distinguish between Type I and Type II errors
- ❸ It always favors complex models
- ❹ It cannot be used for imbalanced datasets

ANSWER: B

DETAILED TUTORIAL:

• **Limitations of accuracy:**

❶ **Doesn't distinguish error types:**

- FP and FN treated equally
- But costs are usually different

❷ **Misleading with imbalanced data:**

- 98% positive, classify all as positive
- Accuracy = 98%, but model is useless

❸ **Problematic example:**

- Fraud detection: 99% legitimate
- Model predicting "legitimate" always has 99% accuracy
- But catches zero fraud (Recall = 0)

EXAM TRAP: Accuracy is misleading for imbalanced datasets!

QUESTION: What is precision in classification model evaluation?

OPTIONS:

- ❶ Of all actual positives, how many were correctly identified?
- ❷ Of all predicted positives, how many were actually positive?
- ❸ Of all predictions, how many were correct?
- ❹ Average of true positive and true negative rates

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

$$\bullet \text{ Precision} = \frac{TP}{TP+FP}$$

• **Managerial question:**

- "Of all the alerts our model generated, how many were actually real issues?"
- "If we act on this prediction, how often will we be right?"

• **When precision matters:**

- High cost of investigation (AML alerts)
- Customer friction concerns
- Operational resource constraints

EXAM TRAP: Precision = $TP / (TP + FP)$ - "of predicted positives, how many were right?"

Q37: Precision Example - Tutorial

QUESTION: In the dividend example with $TP=432$, $FP=121$, what is precision?

OPTIONS:

- 1 72.0%
- 2 78.1%
- 3 43.2%
- 4 71.1%

ANSWER: B

DETAILED TUTORIAL:

• **Calculation:**

- $\text{Precision} = 432 / (432 + 121)$
- $= 432 / 553$
- $= 78.1\%$

• **Interpretation:**

- "When the model predicts a firm will pay a dividend, it's right 78.1% of the time"
- "Of all dividend predictions, 78.1% were correct"

• **Context:**

- 432 firms correctly predicted to pay dividends
- 121 firms incorrectly predicted to pay dividends
- Total predicted positive = 553

EXAM TRAP: Precision = $TP / (TP + FP)$!

QUESTION: What is recall (sensitivity) in classification model evaluation?

OPTIONS:

- ❶ Of all predicted positives, how many were actually positive?
- ❷ Of all actual positives, how many were correctly identified?
- ❸ Of all predictions, how many were correct?
- ❹ Average of true positive and true negative rates

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

$$\bullet \text{ Recall} = \frac{TP}{TP+FN}$$

• **Also known as:** Sensitivity or True Positive Rate

• **Managerial question:**

- "Of all real fraud events that occurred, how many did we catch?"
- "What percentage of actual positives did we identify?"

• **When recall matters:**

- Missing a positive has severe consequences
- Fraud detection, default prediction
- Compliance monitoring

EXAM TRAP: Recall = $TP / (TP + FN)$ - "of actual positives, how many did we catch?"

QUESTION: In the dividend example with TP=432, FN=168, what is recall?

OPTIONS:

- 1 72.0%
- 2 78.1%
- 3 43.2%
- 4 71.1%

ANSWER: A

DETAILED TUTORIAL:

• **Calculation:**

- $\text{Recall} = 432 / (432 + 168)$
- $= 432 / 600$
- $= 72.0\%$

• **Interpretation:**

- "Of all firms that actually paid dividends, the model caught 72%"
- "The model missed 28% of dividend payers"

• **Context:**

- 600 firms actually paid dividends
- 432 were correctly predicted
- 168 were missed (False Negatives)

EXAM TRAP: Recall = TP / (TP + FN)!

Q40: Precision vs Recall Tradeoff - Tutorial

QUESTION: What is the relationship between precision and recall?

OPTIONS:

- ① They always move in the same direction
- ② There is often a tradeoff - improving one can reduce the other
- ③ They are independent of each other
- ④ Precision is always higher than recall

ANSWER: B

DETAILED TUTORIAL:

- **The tradeoff:**
 - **High threshold:** Higher precision, lower recall
 - **Low threshold:** Lower precision, higher recall
- **Example:**
 - Threshold = 0.9: Only very confident predictions
 - Precision: High (few false positives)
 - Recall: Low (miss many actual positives)
- **Balancing act:**
 - Business needs determine the balance
 - F1 score captures both
 - Cost matrix helps choose optimal point

EXAM TRAP: Precision and Recall have **tradeoff** - cannot maximize both simultaneously!

QUESTION: What is the F1 Score and why is it useful?

OPTIONS:

- ① Sum of precision and recall
- ② Harmonic mean of precision and recall; balances both in a single metric
- ③ Product of precision and recall
- ④ Average of precision and recall

ANSWER: B

DETAILED TUTORIAL:

- **Formula:**
 - $$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$
- **Why harmonic mean:**
 - Harmonic mean penalizes imbalance
 - Low precision or low recall $\rightarrow$ Low F1
 - Better than arithmetic mean for this purpose
- **Interpretation:**
 - Bounded between 0 and 1
 - Close to 1 = high precision and high recall
 - Low score = poor precision, poor recall, or both

EXAM TRAP: F1 = harmonic mean of precision and recall!

QUESTION: Given Precision=78.1% and Recall=72.0%, what is the F1 Score?

OPTIONS:

- 1 75.0%
- 2 74.9%
- 3 78.1%
- 4 72.0%

ANSWER: B

DETAILED TUTORIAL:

• **Calculation:**

- $F1 = 2 \times (0.781 \times 0.720) / (0.781 + 0.720)$
- $= 2 \times 0.5623 / 1.501$
- $= 1.1246 / 1.501$
- $= 0.7494$ equal 74.9%

• **Interpretation:**

- $F1 = 74.9\%$
- Balances precision and recall
- Good overall performance

EXAM TRAP: F1 equal $2 \times (P \times R) / (P + R)$!

QUESTION: What is specificity (true negative rate)?

OPTIONS:

- ❶ Proportion of predicted positives that are actually positive
- ❷ Proportion of actual negatives correctly predicted as negative
- ❸ Proportion of actual positives correctly predicted as positive
- ❹ Proportion of predictions that are correct

ANSWER: B

DETAILED TUTORIAL:

• **Formula:**

- $Specificity = \frac{TN}{TN+FP}$

• **Interpretation:**

- "Of all actual negatives, what percentage did we correctly identify?"
- True Negative Rate

• **Relation to Recall:**

- Recall = TPR (True Positive Rate)
- Specificity = TNR (True Negative Rate)
- Both are important for ROC curve

EXAM TRAP: Specificity = $TN / (TN + FP)$ - True Negative Rate!

Q44: Imbalanced Data Problem - Tutorial

QUESTION: Why is accuracy misleading for imbalanced datasets?

OPTIONS:

- ❶ Accuracy is too difficult to calculate
- ❷ A model can have high accuracy by always predicting the majority class
- ❸ Accuracy treats all errors equally
- ❹ Accuracy is not a valid metric

ANSWER: B

DETAILED TUTORIAL:

- **The problem:**
 - Dataset: 98% solvent, 2% default
 - Model predicts "solvent" for everyone
 - Accuracy = 98% (seems great!)
 - But model catches zero defaults
- **What metrics to use instead:**
 - Precision: 0% (no defaults predicted)
 - Recall: 0% (caught zero actual defaults)
 - F1: 0% (poor precision and recall)
- **Lesson:**
 - Accuracy is insufficient for imbalanced data
 - Use Precision, Recall, F1, AUC

EXAM TRAP: Accuracy is misleading for imbalanced data!

QUESTION: Given a loan default model with TP=60, TN=498, FP=98, FN=5, what are the performance metrics?

OPTIONS:

- ① Accuracy=84.4%, Precision=38.0%, Recall=92.3%
- ② Accuracy=92.3%, Precision=38.0%, Recall=84.4%
- ③ Accuracy=84.4%, Precision=92.3%, Recall=38.0%
- ④ Accuracy=38.0%, Precision=84.4%, Recall=92.3%

ANSWER: A

DETAILED TUTORIAL:

• **Calculations:**

- Total = $60 + 498 + 98 + 5 = 661$
- Accuracy = $(60 + 498) / 661 = 558/661 = 84.4\%$
- Precision = $60 / (60 + 98) = 60/158 = 38.0\%$
- Recall = $60 / (60 + 5) = 60/65 = 92.3\%$

• **Interpretation:**

- Model catches 92.3% of defaults (great!)
- But only 38% of default predictions are correct
- Many false alarms (98 false positives)

EXAM TRAP: Work through each formula carefully - don't mix up denominators!

QUESTION: What is a Receiver Operating Characteristic (ROC) curve?

OPTIONS:

- ① A plot of model parameters
- ② A graphical illustration of the tradeoff between true positive and false positive rates across different thresholds
- ③ A plot of training vs test performance
- ④ A plot of precision vs recall

ANSWER: B

DETAILED TUTORIAL:

- **ROC Curve:**
 - Y-axis: True Positive Rate (Recall)
 - X-axis: False Positive Rate (1 - Specificity)
 - Points from varying decision threshold
- **Key insight:**
 - Shows tradeoff between catching positives and avoiding false alarms
 - Each threshold gives different TPR and FPR
 - Curve connects all possible thresholds

EXAM TRAP: ROC shows **TPR vs FPR** tradeoff across thresholds!

QUESTION: What does AUC (Area Under the ROC Curve) measure?

OPTIONS:

- ❶ The accuracy of the model at a specific threshold
- ❷ The overall ability of the model to distinguish between positive and negative classes
- ❸ The precision of the model
- ❹ The recall of the model

ANSWER: B

DETAILED TUTORIAL:

- **AUC Interpretation:**
 - $AUC = 1.0$: Perfect discrimination
 - $AUC = 0.5$: No discrimination (random guessing)
 - $AUC < 0.5$: Worse than random (should invert predictions)
 - Higher AUC = better model
- **Probabilistic interpretation:**
 - Probability that a random positive ranks higher than a random negative
 - Measures ranking ability
 - Threshold-independent

EXAM TRAP: AUC = probability that positive ranks higher than negative!

QUESTION: What does an AUC of 0.5 indicate about a model?

OPTIONS:

- ① Perfect predictive ability
- ② No predictive ability (random predictions)
- ③ Worse than random predictions
- ④ The model is overfitting

ANSWER: B

DETAILED TUTORIAL:

- **AUC values and interpretation:**
 - AUC = 1.0: Perfect model
 - AUC = 0.5: Random model (no predictive ability)
 - AUC $\downarrow$ 0.5: Worse than random (should invert)
 - Typical good models: 0.7-0.9
- **What AUC = 0.5 means:**
 - Model cannot distinguish between classes
 - Equivalent to random guessing
 - No useful predictive information

EXAM TRAP: AUC = 0.5 = no predictive ability - random guessing!

QUESTION: What does an AUC of 1.0 indicate about a model?

OPTIONS:

- ❶ No predictive ability
- ❷ Perfect predictive ability (all positives rank above all negatives)
- ❸ Worse than random
- ❹ The model is overfitting

ANSWER: B

DETAILED TUTORIAL:

- **AUC = 1.0 meaning:**
 - Perfect separation between classes
 - Every positive has higher score than every negative
 - There exists a threshold with perfect classification
 - Usually indicates overfitting or data leakage
- **Caution:**
 - AUC = 1.0 on test data is suspicious
 - Likely indicates problem with data or methodology
 - Real-world data rarely has AUC = 1.0

EXAM TRAP: AUC = 1.0 = perfect separation - be suspicious!

QUESTION: Which model would have a better ROC curve?

OPTIONS:

- ① Model with curve closer to the upper-left corner (higher TPR, lower FPR)
- ② Model with curve closer to the diagonal line
- ③ Model with curve closer to the lower-right corner
- ④ All ROC curves are equally good

ANSWER: A

DETAILED TUTORIAL:

- **ROC curve interpretation:**
 - Upper-left corner: $TPR=1$, $FPR=0$ (perfect)
 - Diagonal line: $TPR=FPR$ (random)
 - Closer to upper-left = better model
- **Comparison:**
 - Curve A with more area = better
 - Curve that stays above and left = better
 - AUC captures this visually

EXAM TRAP: Better ROC curve = closer to upper-left corner!

Q51: Threshold Effect on ROC - Tutorial

QUESTION: What happens to ROC points as threshold changes from 0 to 1?

OPTIONS:

- ❶ Points move from (0,0) to (1,1) along the ROC curve
- ❷ Points move from (1,1) to (0,0) along the ROC curve
- ❸ Points move randomly
- ❹ Points stay in one place

ANSWER: A

DETAILED TUTORIAL:

- **Threshold extremes:**
 - **Threshold = 0:** All predicted positive
 - $TPR = 1$ (all positives caught)
 - $FPR = 1$ (all negatives misclassified)
 - Point: (1, 1)
 - **Threshold = 1:** All predicted negative
 - $TPR = 0$ (no positives caught)
 - $FPR = 0$ (all negatives correct)
 - Point: (0, 0)
- **As threshold increases:**
 - Move from (1,1) toward (0,0)
 - TPR decreases (catch fewer positives)
 - FPR decreases (fewer false alarms)

EXAM TRAP: ROC curve goes from (0,0) to (1,1) as threshold decreases!

QUESTION: How can AUC be used to compare models in lending decisions?

OPTIONS:

- ① Higher AUC indicates better ability to separate good from bad loans
- ② Lower AUC indicates better performance
- ③ AUC is not useful for model comparison
- ④ Models with same AUC perform identically

ANSWER: A

DETAILED TUTORIAL:

- **AUC in lending:**
 - Measures ability to rank borrowers by risk
 - Higher AUC = better separation
 - Better model = more accurate risk ranking
- **Decision context:**
 - Choose model with higher AUC on test set
 - AUC = 0.8 vs 0.7: First has better ranking
 - Better AUC = more confidence in decisions

EXAM TRAP: Higher AUC = **better model** for lending decisions!

QUESTION: When would you use a Precision-Recall curve instead of ROC?

OPTIONS:

- ① When the dataset is perfectly balanced
- ② When the dataset is highly imbalanced (very few positives)
- ③ When you only care about accuracy
- ④ When you don't need to visualize performance

ANSWER: B

DETAILED TUTORIAL:

- **Precision-Recall curves:**
 - Better for imbalanced datasets
 - Focus on positive class performance
 - More sensitive to rare events
- **Why not ROC:**
 - ROC can be optimistic with imbalance
 - Large number of negatives makes FPR low
 - AUC can be high even with poor positive performance

EXAM TRAP: Precision-Recall for **imbalanced data**!

QUESTION: How are classification metrics extended to multi-class problems?

OPTIONS:

- ① Use the same formulas, just sum all classes
- ② Compute metrics for each class vs all others (one-vs-rest)
- ③ Multi-class problems cannot be evaluated
- ④ Use only accuracy for multi-class

ANSWER: B

DETAILED TUTORIAL:

- **One-vs-rest approach:**
 - For each class, treat as "positive"
 - All other classes as "negative"
 - Compute TP, FP, FN, TN for each class
 - Average or aggregate metrics
- **Example - Credit ratings:**
 - AAA, AA, A, BBB, etc.
 - For AAA: positive = AAA, negative = all others
 - Compute precision/recall for AAA
 - Repeat for each rating

EXAM TRAP: Multi-class uses **one-vs-rest** approach!

QUESTION: What is an advantage of AUC over accuracy?

OPTIONS:

- ① AUC is easier to calculate
- ② AUC is threshold-independent and works well for imbalanced data
- ③ AUC always gives higher values
- ④ AUC requires less data

ANSWER: B

DETAILED TUTORIAL:

- **AUC advantages:**
 - **Threshold-independent:** Evaluates all thresholds
 - **Works with imbalance:** Not skewed by majority class
 - **Ranking ability:** Measures how well model orders cases
- **Accuracy limitations:**
 - Threshold-dependent (uses one threshold)
 - Misleading for imbalanced data
 - Doesn't capture ranking ability

EXAM TRAP: AUC is **threshold-independent** - a key advantage!

QUESTION: What is a decision threshold in classification?

OPTIONS:

- ❶ The point at which training stops
- ❷ A cutoff value applied to predicted probabilities to make binary classifications
- ❸ The learning rate in gradient descent
- ❹ The regularization parameter

ANSWER: B

DETAILED TUTORIAL:

- **Definition:**
 - Most classification models output probabilities
 - Threshold converts probability to class label
 - If $PD \geq \text{threshold} \rightarrow \text{"default"}$
 - If $PD < \text{threshold} \rightarrow \text{"no default"}$
- **Common choices:**
 - Default: 0.5 (balanced)
 - Can be adjusted based on business needs
 - Different thresholds = different tradeoffs

EXAM TRAP: Threshold converts **probability to class label**!

QUESTION: What is a cost matrix in threshold selection?

OPTIONS:

- ❶ A matrix showing model parameters
- ❷ Explicitly defined economic consequences for each quadrant of the confusion matrix
- ❸ A matrix showing feature correlations
- ❹ A matrix showing error rates

ANSWER: B

DETAILED TUTORIAL:

- **Cost Matrix components:**
 - **False Negative:** e.g., loss from undetected fraud
 - **False Positive:** e.g., cost to investigate alert
 - **True Positive:** e.g., benefit of catching fraud
 - **True Negative:** e.g., benefit of smooth processing
- **Why it matters:**
 - Different costs = different optimal thresholds
 - Balances business economics with model decisions
 - Makes threshold selection transparent

EXAM TRAP: Cost matrix = **economic consequences** of each decision!

QUESTION: Who should be involved in developing a cost matrix?

OPTIONS:

- ❶ Only the modeling team
- ❷ Business lines, operations, finance, risk, compliance, and legal
- ❸ Only senior management
- ❹ Only regulators

ANSWER: B

DETAILED TUTORIAL:

- **Key stakeholders:**
 - **Business lines:** Cost of missed opportunities, customer attrition
 - **Operations:** Cost of manual reviews
 - **Finance/Risk:** Financial impact of missed risks
 - **Compliance/Legal:** Regulatory breach costs
- **Why multiple perspectives:**
 - Different stakeholders understand different costs
 - Complete picture of all consequences
 - Collaborative decisions are more defensible

EXAM TRAP: Cost matrix needs **cross-functional input**!

Q59: Dynamic Threshold Adjustment - Tutorial

QUESTION: Should thresholds be static or dynamic?

OPTIONS:

- ① Always static (set once and never change)
- ② Should be dynamic and adjusted based on changing conditions and risk appetite
- ③ Always use 0.5
- ④ Thresholds cannot be changed

ANSWER: B

DETAILED TUTORIAL:

- **Dynamic adjustment examples:**
 - **Holiday fraud spike:** Lower threshold (higher recall)
 - **Economic downturn:** Adjust default thresholds
 - **Benign periods:** Higher threshold (fewer false positives)
- **Prerequisites:**
 - Clear risk appetite statement
 - Documented review process
 - Transparent justification
 - Regulatory awareness

EXAM TRAP: Thresholds should be **dynamic** - not "set and forget"!

QUESTION: Why should threshold choices be documented and defensible?

OPTIONS:

- ① Because it's a regulatory requirement
- ② To justify decisions to senior management, board, and regulators
- ③ Because documentation is standard practice
- ④ All of the above

ANSWER: D

DETAILED TUTORIAL:

- **Documentation should show:**
 - How threshold aligns with risk appetite
 - Tradeoff between recall and false positives
 - Costs and benefits considered
 - Periodic review process
- **Example statement:**
 - "Threshold set for AML monitoring to achieve 80% recall"
 - "Acceptable false positive rate of X%"
 - "Deemed acceptable given consequences of missing AML events"

EXAM TRAP: Threshold choices must be **transparent and defensible!**

QUESTION: What costs are associated with False Positives in AML monitoring?

OPTIONS:

- ① No costs
- ② Operational cost to investigate, customer friction, resource waste
- ③ Only financial costs
- ④ Only reputational costs

ANSWER: B

DETAILED TUTORIAL:

- **False Positive costs:**
 - ① **Operational:** Investigator time wasted
 - ② **Customer:** Friction, declined legitimate transactions
 - ③ **Resource:** Compliance teams chasing red herrings
 - ④ **Desensitization:** Alert fatigue, missing real issues
- **Why they matter:**
 - High FP rate = inefficient operations
 - Can outweigh benefits of catching fraud
 - Affects customer experience

EXAM TRAP: False Positives = operational and customer costs!

QUESTION: What costs are associated with False Negatives in default prediction?

OPTIONS:

- ① No costs
- ② Financial losses, regulatory capital implications, missed risk
- ③ Only reputational costs
- ④ Only operational costs

ANSWER: B

DETAILED TUTORIAL:

- **False Negative costs:**
 - ① **Financial:** Actual loan losses
 - ② **Regulatory:** Capital implications
 - ③ **Risk:** Misclassification of risk
 - ④ **Reputational:** Regulatory sanctions
- **Why they're often the largest cost:**
 - Direct financial impact
 - Hard to recover from
 - Regulatory consequences

EXAM TRAP: False Negatives = financial and regulatory costs - usually the largest!

QUESTION: How is the optimal threshold determined using a cost matrix?

OPTIONS:

- ① Always use 0.5
- ② Use the threshold that minimizes total expected cost
- ③ Use the threshold that maximizes accuracy
- ④ Use the threshold that maximizes AUC

ANSWER: B

DETAILED TUTORIAL:

- **Expected cost calculation:**
 - For each threshold, compute expected cost
 - $\text{Cost} = C(\text{TP}) \times P(\text{TP}) + C(\text{FP}) \times P(\text{FP}) + \dots$
 - Choose threshold with minimum cost
- **Cost-based threshold:**
 - Different from accuracy-based threshold
 - Reflects business reality
 - More defensible

EXAM TRAP: Optimal threshold = minimum expected cost!

QUESTION: Which model would be preferred for fraud detection?

OPTIONS:

- ❶ Model with high precision but low recall
- ❷ Model with high recall (catches most fraud) even if precision is lower
- ❸ Model with equal precision and recall
- ❹ Model with highest accuracy

ANSWER: B

DETAILED TUTORIAL:

- **Fraud detection priorities:**
 - Missing fraud (False Negative) is very costly
 - Better to catch more fraud (high recall)
 - Accept some false alarms (lower precision)
 - Business needs drive the tradeoff
- **Other contexts:**
 - AML monitoring: High recall critical
 - Credit approval: Balance needed
 - Marketing: Precision more important

EXAM TRAP: Fraud detection prioritizes **recall** over **precision**!

QUESTION: How often should model thresholds be reviewed?

OPTIONS:

- ① Never - set them once
- ② Periodically, based on changing conditions and risk environment
- ③ Only when regulators ask
- ④ Annually without exception

ANSWER: B

DETAILED TUTORIAL:

- **Review triggers:**
 - Significant environmental changes
 - Shifts in risk appetite
 - New regulatory requirements
 - Changes in fraud patterns
- **Best practice:**
 - Documented review schedule
 - Clear triggers for review
 - Stakeholder involvement
 - Regulatory notification when appropriate

EXAM TRAP: Thresholds should be **periodically reviewed** - not static!

Q66: Model Comparison Example - Tutorial

QUESTION: In the LendingClub example, how do logistic regression and neural network compare?

OPTIONS:

- ❶ Logistic regression is clearly superior
- ❷ Neural network is clearly superior
- ❸ Different metrics give different rankings; choice depends on priorities
- ❹ Both models perform identically on all metrics

ANSWER: C

DETAILED TUTORIAL:

- **Test sample comparison (Table 8.6):**
 - Accuracy: LR=71.3%, NN=70.1% (LR better)
 - Precision: LR=60.0%, NN=54.1% (LR better)
 - Recall: LR=28.3%, NN=37.7% (NN better)
- **Interpretation:**
 - No clear winner
 - Logistic regression: More accurate, better precision
 - Neural network: Catches more defaults (better recall)
 - Choice depends on business priorities

EXAM TRAP: Different metrics = **different model rankings!** Choose based on problem.

Q67: Training vs Test Performance Comparison - Tutorial

QUESTION: In the LendingClub example, what does the drop from training to test performance indicate?

OPTIONS:

- ❶ The model is perfectly specified
- ❷ Some overfitting is present (performance drops on unseen data)
- ❸ The data is perfectly clean
- ❹ The model needs more complexity

ANSWER: B

DETAILED TUTORIAL:

- **Performance drops:**
 - Training accuracy: 84.2% → Test: 71.3% (LR)
 - Training accuracy: 84.2% → Test: 70.1% (NN)
 - Both show decrease
- **Interpretation:**
 - Models overfit to training data
 - Reasonable amount of overfitting
 - Could improve with regularization
 - Remove irrelevant features

EXAM TRAP: Performance drop = **overfitting** - expected but should be managed!

QUESTION: How can you check for overfitting in neural networks?

OPTIONS:

- ① Look at training vs test performance gap
- ② Examine weights for very strong or offsetting connections
- ③ Both A and B
- ④ Neural networks cannot overfit

ANSWER: C

DETAILED TUTORIAL:

- **Overfitting checks:**
 - ① **Performance:** Training $\downarrow$ Test
 - ② **Weights:** Very large or offsetting values
 - ③ **Loss:** Training loss flat, validation increasing
- **What to look for:**
 - Large positive and negative weights
 - Model memorizing training data
 - Poor generalization

EXAM TRAP: Check both performance gap and weight patterns for overfitting!

QUESTION: What are advantages of logistic regression over neural networks?

OPTIONS:

- ① Always higher accuracy
- ② More interpretable, easier to explain, faster to train
- ③ More complex and powerful
- ④ Requires less data

ANSWER: B

DETAILED TUTORIAL:

- **Logistic regression advantages:**
 - **Interpretability:** Coefficients explain feature impact
 - **Explainability:** Easy to communicate to stakeholders
 - **Speed:** Faster to train and deploy
 - **Transparency:** Regulatory acceptance
- **Neural network advantages:**
 - Can capture complex patterns
 - Higher potential accuracy
 - Better for large datasets

EXAM TRAP: Logistic regression = **more interpretable** than neural networks!

QUESTION: What are advantages of neural networks over logistic regression?

OPTIONS:

- ① Always more interpretable
- ② Can capture complex nonlinear patterns, better for some problems
- ③ Faster to train
- ④ Requires less data

ANSWER: B

DETAILED TUTORIAL:

- **Neural network advantages:**
 - **Complex patterns:** Can model nonlinear relationships
 - **Flexibility:** Universal approximator
 - **Performance:** Often better for complex tasks
 - **Feature learning:** Learns representations automatically
- **Disadvantages:**
 - Less interpretable
 - More data needed
 - More computationally expensive
 - Risk of overfitting

EXAM TRAP: Neural networks = better for complex patterns but less interpretable!

QUESTION: In the LendingClub example, which model would a risk manager choose?

OPTIONS:

- ① Always choose logistic regression
- ② Always choose neural network
- ③ Depends on priorities: recall vs precision, interpretability vs accuracy
- ④ Choose the model with highest accuracy

ANSWER: C

DETAILED TUTORIAL:

- **Tradeoffs:**
 - **If catching defaults is critical:** Neural network (higher recall)
 - **If interpretability is required:** Logistic regression
 - **If accuracy is priority:** Logistic regression (slightly higher)
 - **If regulatory compliance:** Logistic regression
- **Key insight:**
 - No single "best" model
 - Choice depends on business needs
 - Must align with risk appetite

EXAM TRAP: Model choice = depends on priorities and constraints!

QUESTION: How can overfitting be addressed in the LendingClub example?

OPTIONS:

- ① Add more complexity to the model
- ② Remove irrelevant features or apply regularization
- ③ Use more training data only
- ④ Nothing - overfitting is unavoidable

ANSWER: B

DETAILED TUTORIAL:

- Remediation strategies:
 - ① **Simplify model:** Remove irrelevant features
 - ② **Add regularization:** L1, L2, dropout
 - ③ **Get more data:** More training samples
 - ④ **Cross-validation:** Better model selection
 - ⑤ **Early stopping:** Stop before overfitting
- Chapter recommendation:
 - "Remove some of the least empirically relevant features"
 - "Apply regularization to the fitted models"

EXAM TRAP: Fix overfitting with [simplification](#) or [regularization](#)!

QUESTION: Which metric should be used to compare models for default prediction?

OPTIONS:

- ① Only accuracy
- ② A combination of metrics based on business priorities
- ③ Only precision
- ④ Only recall

ANSWER: B

DETAILED TUTORIAL:

- **Recommended approach:**
 - ① Start with confusion matrix
 - ② Compute Accuracy, Precision, Recall, F1
 - ③ Consider business costs
 - ④ Choose metric that aligns with priorities
- **Why multiple metrics:**
 - Single metric masks important differences
 - Different stakeholders need different views
 - Regulatory requirements may specify

EXAM TRAP: Use **multiple metrics** - don't rely on just one!

QUESTION: Why is the test sample comparison more important than training sample comparison?

OPTIONS:

- ❶ Test sample is larger
- ❷ Test sample shows true generalization performance
- ❸ Training sample is always more accurate
- ❹ Test sample is easier to compute

ANSWER: B

DETAILED TUTORIAL:

- **Training sample:**
 - Shows how well model memorizes
 - Optimistically biased
 - Not representative of new data
- **Test sample:**
 - Shows how well model generalizes
 - Unbiased performance estimate
 - What matters in practice

EXAM TRAP: Test sample = true generalization - always prioritize!

QUESTION: What is the tradeoff between logistic regression and neural networks?

OPTIONS:

- ❶ Logistic regression is more complex
- ❷ Logistic regression is more interpretable but may have lower accuracy
- ❸ Neural networks are always better
- ❹ There is no tradeoff

ANSWER: B

DETAILED TUTORIAL:

- **Logistic regression:**
 - **Pros:** Interpretable, transparent, regulatory friendly
 - **Cons:** Limited to linear relationships
- **Neural networks:**
 - **Pros:** Can capture complex patterns
 - **Cons:** Black box, hard to explain
- **Choice depends on:**
 - Regulatory requirements
 - Business needs
 - Stakeholder expectations

EXAM TRAP: Tradeoff = [interpretability](#) vs [performance](#)!

QUESTION: What is the key takeaway from the LendingClub model comparison?

OPTIONS:

- ① Neural networks always outperform logistic regression
- ② Logistic regression always outperforms neural networks
- ③ There is no universal best model; choice depends on evaluation criteria
- ④ Both models perform identically

ANSWER: C

DETAILED TUTORIAL:

• **Key insights:**

- Different metrics gave different rankings
- Logistic regression: Better accuracy and precision
- Neural network: Better recall
- No clear winner

• **Implication:**

- Model choice must align with business objectives
- "Contradictory indicators illustrate importance of choosing appropriate evaluation metric"
- Regulatory context matters

EXAM TRAP: No universal best model - choice depends on criteria!

QUESTION: Why should evaluation metrics align with business objectives?

OPTIONS:

- ❶ Because it's a regulatory requirement
- ❷ To ensure model decisions reflect business priorities and risk appetite
- ❸ Because it makes calculations easier
- ❹ Because all metrics are equally important

ANSWER: B

DETAILED TUTORIAL:

- **Business alignment importance:**
 - Metrics drive model selection
 - Model selection drives decisions
 - Decisions have real-world consequences
 - Wrong metric = wrong decisions
- **Example:**
 - If catching fraud is priority → optimize recall
 - If minimizing investigations → optimize precision
 - Must match business needs

EXAM TRAP: Metrics must align with business objectives!

QUESTION: How do regulatory requirements affect model evaluation?

OPTIONS:

- ❶ Regulators don't care about model evaluation
- ❷ Regulators may specify required metrics and documentation (ECB, regulators)
- ❸ Only internal metrics matter
- ❹ Regulations only apply to training

ANSWER: B

DETAILED TUTORIAL:

- **Regulatory expectations:**
 - ECB recommends tail-risk metrics
 - RMSE alongside MAE
 - Documentation of metric choice
 - Justification for model selection
- **Why it matters:**
 - Compliance is mandatory
 - Penalties for non-compliance
 - Reputational impact

EXAM TRAP: Regulators **specify metric requirements!**

QUESTION: How should multiple models be compared in practice?

OPTIONS:

- ❶ Choose the one with highest accuracy only
- ❷ Use multiple metrics, consider business context, check regulatory requirements
- ❸ Always choose the simplest model
- ❹ Always choose the most complex model

ANSWER: B

DETAILED TUTORIAL:

• Comparison framework:

- ❶ **Multiple metrics:** Accuracy, Precision, Recall, F1, AUC
- ❷ **Business context:** Costs, priorities, risk appetite
- ❸ **Regulatory:** Required metrics, documentation
- ❹ **Interpretability:** Stakeholder needs
- ❺ **Performance:** Training vs. test gap

QUESTION: What is the recommended framework for model performance evaluation?

OPTIONS:

- ① Use one metric for all problems
- ② Use multiple metrics, understand business context, document choices
- ③ Always use the latest metric
- ④ Only use accuracy

ANSWER: B

DETAILED TUTORIAL:

• **Recommended framework:**

- ① **Continuous:** MSE, RMSE, MAE, MAPE
- ② **Classification:** Confusion Matrix, Accuracy, Precision, Recall, F1, AUC
- ③ **Business context:** Cost matrix, risk appetite
- ④ **Documentation:** Transparent and defensible

• **Key principle:**

- No single metric tells the whole story
- Choose based on problem and stakeholders
- Document and justify choices

EXAM TRAP: Comprehensive evaluation = multiple metrics + business context + documentation!

QUESTION: What can error distributions reveal about model performance?

OPTIONS:

- ❶ Nothing beyond average metrics
- ❷ Systematic biases, segments with larger errors, distribution shape
- ❸ Only the average error
- ❹ Only the maximum error

ANSWER: B

DETAILED TUTORIAL:

- **Key insights from distributions:**
 - **Bias:** Systematic over/under prediction
 - **Segments:** Areas where model performs poorly
 - **Shape:** Many small vs few large errors
- **Visualizations to use:**
 - Histograms of residuals
 - Box plots by segment
 - Q-Q plots for normality

EXAM TRAP: Error distributions reveal **biases and patterns** that averages hide!

QUESTION: What does systematic bias in model errors indicate?

OPTIONS:

- ❶ The model is perfect
- ❷ The model consistently over-predicts or under-predicts in certain conditions
- ❸ The data is clean
- ❹ The model is overfitting

ANSWER: B

DETAILED TUTORIAL:

- **Systematic bias examples:**
 - **Over-prediction:** Consistently predicting higher values
 - **Under-prediction:** Consistently predicting lower values
 - **Segment bias:** Errors in specific subgroups
- **Why it matters:**
 - Can lead to systematic business impact
 - Harder to correct than random errors
 - May indicate model misspecification

EXAM TRAP: Systematic bias = consistent error pattern - more problematic than random errors!

QUESTION: Why should performance be analyzed across different segments?

OPTIONS:

- ❶ It's not necessary
- ❷ To identify segments where errors are consistently larger or more biased
- ❸ Because all segments perform identically
- ❹ Because it's a regulatory requirement only

ANSWER: B

DETAILED TUTORIAL:

- **Segment analysis benefits:**
 - Identifies model weaknesses
 - Reveals fairness issues
 - Highlights data quality problems
 - Guides model improvements
- **Example segments:**
 - Geographic regions
 - Product types
 - Customer segments
 - Economic conditions

EXAM TRAP: Segment analysis reveals **where** models fail!

QUESTION: Why do regulators prefer metrics that capture tail risk?

OPTIONS:

- ① Because tail events are rare and don't matter
- ② Because extreme events have severe financial consequences
- ③ Because all errors are equally important
- ④ Because they don't trust MAE

ANSWER: B

DETAILED TUTORIAL:

- **Tail risk importance:**
 - Extreme events cause largest losses
 - Capital requirements driven by tail risk
 - Regulatory focus on solvency
 - Systemic risk concerns
- **Why MSE/RMSE:**
 - Heavily penalize large errors
 - Capture tail events better than MAE
 - Align with regulatory capital frameworks

EXAM TRAP: Regulators want **tail-risk metrics** for capital adequacy!

QUESTION: What should be included in a cost matrix?

OPTIONS:

- ① Only financial costs
- ② All costs: financial, operational, reputational, regulatory
- ③ Only operational costs
- ④ Only regulatory costs

ANSWER: B

DETAILED TUTORIAL:

- **Cost components:**
 - ① **Financial:** Losses from missed risks, capital impact
 - ② **Operational:** Investigation costs, resource waste
 - ③ **Reputational:** Customer trust, market confidence
 - ④ **Regulatory:** Fines, sanctions, scrutiny
- **Benefit components:**
 - **True Positive:** Losses avoided, risks managed
 - **True Negative:** Efficient processing, satisfied customers

EXAM TRAP: Cost matrix should capture **all consequences** - financial and non-financial!

QUESTION: How should threshold choices be justified to regulators?

OPTIONS:

- ❶ "We used the default 0.5"
- ❷ Through documented analysis showing alignment with risk appetite and cost-benefit tradeoffs
- ❸ "The model developer chose it"
- ❹ "We use the same threshold for all models"

ANSWER: B

DETAILED TUTORIAL:

• **Justification should include:**

- ❶ Analysis of tradeoffs
- ❷ Cost-benefit calculations
- ❸ Alignment with risk appetite
- ❹ Stakeholder consultation
- ❺ Periodic review process

• **Example statement:**

- "Threshold chosen to achieve 80% recall given severe consequences of missing fraud events"
- "Accepts X% false positives costing Y in operational overhead"

EXAM TRAP: Thresholds must be justified with documented analysis!

QUESTION: How should model performance be communicated to different stakeholders?

OPTIONS:

- ❶ Use the same metrics for everyone
- ❷ Use appropriate metrics for each audience: MAPE for boards, RMSE for regulators, detailed metrics for model developers
- ❸ Only communicate to senior management
- ❹ Don't communicate model performance

ANSWER: B

DETAILED TUTORIAL:

- **Audience considerations:**
 - **Board/Management:** MAPE, Accuracy (percentage-based)
 - **Regulators:** RMSE with MAE, tail-risk metrics
 - **Model Developers:** All metrics, error distributions
 - **Audit Committee:** MAPE for standardized comparison
- **Communication best practices:**
 - Use visualizations (ROC curves, error distributions)
 - Explain what metrics mean in business terms
 - Highlight tradeoffs and limitations

EXAM TRAP: Different stakeholders need **different metrics and explanations!**

Q88: Performance Monitoring Frequency - Tutorial

QUESTION: How often should model performance be monitored?

OPTIONS:

- ➊ Only during model development
- ➋ Continuously or at regular intervals (e.g., quarterly, annually)
- ➌ Only when regulators ask
- ➍ Never after deployment

ANSWER: B

DETAILED TUTORIAL:

- **Monitoring frequency:**
 - **Regular intervals:** Quarterly, semi-annually, annually
 - **Event-driven:** Significant environmental changes
 - **Trigger-based:** Performance threshold breaches
- **What to monitor:**
 - Performance metrics over time
 - Error distributions
 - Segment performance
 - Threshold appropriateness

EXAM TRAP: Model performance must be **monitored continuously** - not just during development!

QUESTION: What is the role of validation in model performance evaluation?

OPTIONS:

- ❶ Validation is the same as training
- ❷ Validation provides independent assessment of model performance
- ❸ Validation is only for data preparation
- ❹ Validation is not necessary

ANSWER: B

DETAILED TUTORIAL:

- **Validation purposes:**
 - ❶ Independent performance assessment
 - ❷ Hyperparameter tuning
 - ❸ Overfitting detection
 - ❹ Model comparison
- **Validation best practices:**
 - Use separate validation data
 - Use cross-validation
 - Document validation process
 - Regulatory compliance

EXAM TRAP: Validation provides **independent assessment** of model performance!

QUESTION: Why should performance metric choices be documented?

OPTIONS:

- ❶ It's not necessary
- ❷ For transparency, regulatory compliance, and model governance
- ❸ Only if regulators ask
- ❹ Because it's standard practice

ANSWER: B

DETAILED TUTORIAL:

• **Documentation should include:**

- ❶ Choice of metrics
- ❷ Justification for choices
- ❸ Business context
- ❹ Regulatory requirements
- ❺ Performance results
- ❻ Model selection rationales

• **Why it matters:**

- Regulatory compliance
- Audit trail
- Knowledge transfer
- Defensible decisions

EXAM TRAP: Metric choices must be **documented and justified!**

QUESTION: How can model performance decay be detected?

OPTIONS:

- ❶ It cannot be detected
- ❷ By monitoring performance metrics over time and comparing to baseline
- ❸ Only when models fail catastrophically
- ❹ By monitoring training performance only

ANSWER: B

DETAILED TUTORIAL:

- **Detection methods:**
 - Track metrics over time
 - Compare to baseline performance
 - Monitor error distributions
 - Segment performance tracking
- **Common causes:**
 - Data drift (concept drift)
 - Economic changes
 - Regulatory changes
 - Model degradation

EXAM TRAP: Monitor metrics over time to detect **performance decay**!

QUESTION: What is concept drift and how does it affect model performance?

OPTIONS:

- ❶ It doesn't affect model performance
- ❷ Changes in the underlying relationship between inputs and outputs over time
- ❸ Only affects training data
- ❹ Only affects test data

ANSWER: B

DETAILED TUTORIAL:

- **Concept drift examples:**
 - Changing fraud patterns
 - Economic cycles affecting default relationships
 - Customer behavior changes
 - Regulatory changes
- **Detection and response:**
 - Monitor performance metrics
 - Regularly retrain models
 - Update model parameters
 - Review thresholds

EXAM TRAP: Concept drift = **changing relationships** over time - requires monitoring!

Q93: Model Retraining Triggers - Tutorial

QUESTION: What should trigger model retraining?

OPTIONS:

- ① Never - models are permanent
- ② Performance decay below thresholds, significant data changes, or scheduled intervals
- ③ Only when regulators require
- ④ Randomly

ANSWER: B

DETAILED TUTORIAL:

- Retraining triggers:
 - ① **Performance:** Metrics below acceptable levels
 - ② **Data:** Significant changes in data distribution
 - ③ **Schedule:** Regular intervals (monthly, quarterly)
 - ④ **Events:** Major economic or regulatory changes
- Best practice:
 - Document retraining triggers
 - Regular performance review
 - Proactive monitoring
 - Automated alerts

EXAM TRAP: Retrain when performance decays, data changes, or on schedule!

QUESTION: What should be included in model performance reports?

OPTIONS:

- ❶ Only the latest metrics
- ❷ Current metrics, trends, comparisons to baseline, and any issues identified
- ❸ Only when performance is poor
- ❹ Only for regulatory submissions

ANSWER: B

DETAILED TUTORIAL:

• **Report components:**

- ❶ **Current:** Latest performance metrics
- ❷ **Trends:** Performance over time
- ❸ **Baseline:** Comparison to initial/expected performance
- ❹ **Issues:** Any identified problems or concerns
- ❺ **Actions:** Planned or taken remediation

• **Audience:**

- Model owners
- Management
- Regulators
- Audit committees

EXAM TRAP: Reports should include **current metrics, trends, and issues!**

QUESTION: How does model performance evaluation help identify fairness issues?

OPTIONS:

- ❶ It doesn't - fairness is separate
- ❷ By analyzing performance across different demographic or customer segments
- ❸ Only if fairness metrics are explicitly included
- ❹ Fairness cannot be measured

ANSWER: B

DETAILED TUTORIAL:

- **Fairness evaluation:**
 - Compare performance across segments
 - Check for systematic bias
 - Ensure equal treatment
 - Regulatory compliance (fair lending)
- **What to examine:**
 - Accuracy differences
 - False positive/negative rates
 - Disparate impact
 - Equal opportunity

EXAM TRAP: Segment analysis reveals fairness issues!

QUESTION: How does model performance evaluation relate to explainable AI?

OPTIONS:

- ❶ It doesn't - they are unrelated
- ❷ Performance evaluation identifies where models need better explanations
- ❸ XAI is only for regulators
- ❹ XAI replaces performance evaluation

ANSWER: B

DETAILED TUTORIAL:

- **Connection to XAI:**
 - Poor performance in segments → need explanations
 - Complex models (DNNs) → need XAI techniques
 - Regulatory requirements for explanation
 - Tools: SHAP, LIME for interpretation
- **Practical application:**
 - Use explanations to understand poor performance
 - Identify feature importance
 - Debug model issues
 - Build trust in predictions

EXAM TRAP: Performance evaluation guides **where explanations are needed!**

QUESTION: How do SHAP values help with model performance evaluation?

OPTIONS:

- ❶ They don't - they only explain predictions
- ❷ They help identify which features are driving good and bad performance
- ❸ They replace performance metrics
- ❹ They only work for linear models

ANSWER: B

DETAILED TUTORIAL:

- **SHAP in evaluation:**
 - Identify important features
 - Understand model behavior
 - Debug performance issues
 - Explain to stakeholders
- **Application:**
 - Feature importance analysis
 - Understanding model decisions
 - Regulatory explanations
 - Model improvement insights

EXAM TRAP: SHAP helps understand what drives model performance!

QUESTION: When is LIME particularly useful for model evaluation?

OPTIONS:

- ❶ For evaluating overall model performance
- ❷ For explaining individual predictions and understanding edge cases
- ❸ For model training
- ❹ For data preparation

ANSWER: B

DETAILED TUTORIAL:

- **LIME applications:**
 - Explain specific predictions
 - Understand why model made certain decisions
 - Identify problematic cases
 - Build trust with stakeholders
- **Evaluation insights:**
 - Why certain cases were misclassified
 - What features drove errors
 - Edge case behavior
 - Model limitations

EXAM TRAP: LIME explains **individual predictions** - useful for error analysis!

QUESTION: What is the role of model governance in performance evaluation?

OPTIONS:

- ① No role - models govern themselves
- ② Ensures consistent evaluation, documentation, and monitoring across all models
- ③ Only for regulatory compliance
- ④ Only for complex models

ANSWER: B

DETAILED TUTORIAL:

- **Governance components:**
 - ① **Standardization:** Consistent metrics across models
 - ② **Documentation:** Clear records of evaluation
 - ③ **Monitoring:** Regular performance review
 - ④ **Oversight:** Independent validation
- **Why it matters:**
 - Regulatory compliance
 - Risk management
 - Consistent decision-making
 - Auditability

EXAM TRAP: Governance ensures consistent evaluation and monitoring!

QUESTION: What are the key takeaways from model performance evaluation?

OPTIONS:

- ❶ Only accuracy matters
- ❷ Use multiple metrics, consider business context, monitor continuously, document thoroughly
- ❸ Only use what regulators require
- ❹ Evaluation is only needed during development

ANSWER: B

DETAILED TUTORIAL:

• **Key takeaways:**

- ❶ **Multiple metrics:** Use continuous (MSE, RMSE, MAE, MAPE) and classification (Confusion Matrix, Accuracy, Precision, Recall, F1, AUC)
- ❷ **Business context:** Align with business objectives and risk appetite
- ❸ **Continuous monitoring:** Track performance over time
- ❹ **Documentation:** Record and justify all choices

• **Critical insights:**

- No single metric tells the whole story
- Different metrics serve different purposes
- Performance evaluation is a continuous process
- Regulatory requirements must be met

EXAM TRAP: Comprehensive evaluation = multiple metrics + business context + monitoring + documentation!

Summary: Complete Review

Continuous Output Metrics:

- **MSE:** Average squared error; penalizes large errors; captures tail risk
- **RMSE:** Square root of MSE; same units as target; easier to interpret
- **MAE:** Average absolute error; robust to outliers
- **MAPE:** Percentage error; board-friendly; compares across scales

Classification Metrics:

- **Confusion Matrix:** TP, TN, FP, FN - foundation of all metrics
- **Accuracy:** Total correct / total - misleading for imbalanced data
- **Precision:** $TP / (TP + FP)$ - "of predicted positives, how many were right?"
- **Recall:** $TP / (TP + FN)$ - "of actual positives, how many did we catch?"
- **F1:** Harmonic mean of precision and recall
- **AUC:** Area under ROC curve - threshold-independent ranking ability

Threshold Selection:

- **Cost Matrix:** Economic consequences of each decision
- **Tradeoff:** Balance false positives vs false negatives
- **Dynamic:** Adjust based on changing conditions
- **Documentation:** Transparent and defensible

Exam Success: Understand the metrics, their tradeoffs, and when to use each!